PRE-ALGEBRA (7TH) · LINEAR FUNCTIONS AND GRAPHING

Proportional Relationship Explorer

Explore & Reflect **TEACHER KEY**

Name _____ Date _____ Period _____

Open **Proportional Relationship Explorer** in your unit's Interactive Math section on Mr. K's Classroom (or use your teacher's link). Answer Part 1 **before** you use the tool. Use the tool for Parts 2 and 3. Do Parts 4 and 5 on your own.

1 Predict

before you touch the tool

1. Trail mix costs $2 per pound. Is the total cost proportional to the weight? What if there is also a $3 bag fee?

   Yes: y = 2x. With the fee, no: 2 lb cost $7 but 4 lb cost $11, not $14.

2. Where does the graph of a proportional relationship cross the y-axis?

   At (0, 0).

2 Explore

use the tool

1. Set the price to $2.50 and drag the point to 1 pound. What is its height? Where else does that number appear?

   $2.50. It is k: the price per pound, the change in each row and every value of $\frac{y}{x}$.

2. Add a $4 bag fee. Record $\frac{y}{x}$ at 1, 2 and 4 pounds.

   6.5, 4.5, 3.5: not the same, so not proportional.

3. With the fee, does the cost still go up by the same amount each pound? Is that enough to make it proportional?

   Yes, $2.50 each pound, but it is not proportional because the line doesn't pass through (0, 0).

4. Find a price and a fee that make your store cost the same as Store B ($3 per pound) at 4 pounds.

   Answers vary, e.g., $2 per pound + $4 fee (both cost $12 at 4 lb).

3 Record

use the tool and fill in the table

Compare stores

Price per pound	Bag fee	Equation	Cost of 6 lb	Proportional?
$1.50	$0	y = 1.5x	$9	yes
$2.00	$3	y = 2x + 3	$15	no
$4.00	$0	y = 4x	$24	yes
$0.50	$6	y = 0.5x + 6	$9	no

4 Explain

what did you discover?

1. A relationship is proportional when $\frac{y}{x}$ is **the same (constant)** for every pair.

2. Its graph is a straight line through **(0, 0)** and its equation has the form **y = kx**.

3. The constant of proportionality k is the **unit rate**. It is the value of y when x = **1**.

4. Explain why $y = 2x + 3$ has a straight-line graph but is not proportional.

   Its ratio $\frac{y}{x}$ changes (5, 3.5, 3, …) and it starts at (0, 3), so doubling x does not double y.

---

5 Apply a new situation, no tool

1. A table shows (2, 7), (4, 14), (10, 35). Is it proportional? If so, find k and write the equation.

   Yes. $k = 3.5$, $y = 3.5x$.

2. A table shows (1, 5), (2, 8), (3, 11). Is it proportional? Explain.

   No: $\frac{5}{1} \neq \frac{8}{2}$. (It is $y = 3x + 2$.)

3. A proportional graph passes through (6, 9). Find k and the value of y when $x = 10$.

   $k = 1.5$; $y = 15$.

Teacher notes

Intended discovery: Proportional means a constant ratio $\frac{y}{x} = k$, a line through the origin and an equation $y = kx$; a straight line with a starting value is not proportional.

Likely misconceptions:

- "Every straight line is proportional." (Try this 3 and Explore 3)
- A constant change in the table is enough to be proportional.
- Computing $\frac{y}{x}$ at $x = 0$ (it is undefined).

Answers that can vary: Explore 4: any price p and fee f with $4p + f = 12$ and $p < 3$.